

Industrial Machine Failure Prediction

Abstract

This educational project develops a binary classification prototype for predicting whether an industrial machine observation is associated with failure risk. The workflow includes data preparation, exploratory analysis, preprocessing, Logistic Regression, evaluation, and a Streamlit interface.

Introduction

Unexpected machine failures can interrupt operations and increase maintenance costs. A machine-learning classifier can be used as an analytical prototype to identify patterns associated with failure.

Problem Definition

Given temperature, vibration, pressure, runtime, and maintenance-gap measurements, predict the binary target failure: 0 for no failure risk and 1 for failure risk.

Dataset

The included dataset is synthetic and was generated for this educational prototype. It contains 1,000 observations and five input features plus the target.

Methodology

The workflow uses a train/test split with stratification. Missing values are handled with median imputation, numerical features are standardized, and Logistic Regression is trained as the baseline classifier.

Exploratory Data Analysis

EDA checks structure, missing values, duplicates, descriptive statistics, target distribution, feature distributions, relationships, and potential outliers.

Model Development

The complete preprocessing and Logistic Regression model are stored together in `model.pkl` so the Streamlit application applies the same transformations used during training.

Evaluation

The project reports Accuracy, Precision, Recall, F1-score, and a Confusion Matrix. The values are generated by running `train.py` on the included dataset.

Results

Run train.py to generate the reproducible evaluation metrics for the current dataset and model. Results should be interpreted rather than reported without context.

Limitations

The dataset is synthetic and therefore does not represent real industrial sensor behavior. The model is an educational prototype and must not be used for real maintenance or safety decisions.

Future Scope

Future work could use validated real sensor data, time-aware evaluation, additional models, calibration, monitoring, and domain validation.

Conclusion

The project demonstrates a complete beginner-friendly machine-learning workflow from tabular data preparation to model evaluation and a lightweight prediction application.

References

Learn Depth Academy LLP, Track 1 Final Capstone, Problem 21: Industrial Machine Failure Prediction.
Scikit-learn documentation and Streamlit documentation should be cited in a final academic submission if used.